LEAN STARTUP MANAGEMENT

ASSESSMENT 2

"Police Station Management and Complaint Documentation System"

TEAM MEMBERS:-

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INTRODUCTION:-

The purpose of this assignment is to understand the **real problems faced by police-station personnel** through interviews, identify the most important problem worth solving, and propose a **Minimum Viable Product (MVP)** to address that problem.

The project focuses on improving the process of **complaint documentation and information management** while keeping the police personnel in control of the final decision.

INTERVIEW – UNDERSTANDING USERS AND THEIR PAIN POINTS:-

Target Users

The intended users of the proposed system are:

Target User	Main Responsibility
Police Station Writer	Handles documentation, registers and complaint-related writing/typing
Inspector	Supervises station activities, officers and major cases

The interviews were conducted to understand their **actual work, difficulties and existing methods** rather than starting with a solution.

INTERVIEW FINDINGS – POLICE STATION WRITER

User Role:

POLICE STATION WRITER

Key findings

The Writer explained that complaint-related documentation can take **around one hour** to write and send. The Writer also explained that some complainants do not know how to properly write a complaint, so the Writer has to ask questions and write the complaint for them.

The Writer also stated that there is no dedicated software for some of their station requirements and that they currently write or type the required formats themselves.

Problems identified from the Writer interview

1. Complaint documentation takes considerable time.
2. Some complainants cannot properly write their complaints.
3. The Writer has to ask questions and manually prepare the complaint.
4. Writing and typing creates additional workload.
5. Errors such as spelling mistakes may need correction.
6. There is no dedicated software for some documentation requirements.
7. A system that could capture what the complainant says and assist with documentation was viewed as useful by the Writer.

Important evidence

The Writer stated that it takes about an hour to write and send the complaint-related documentation.

4. INTERVIEW FINDINGS – POLICE INSPECTOR

User Role:

POLICE INSPECTOR

Key Findings

The Inspector explained that managing a police station involves monitoring who is doing what work and following up with officers about their completed and pending work. A duty roster is used, and officers may be contacted to ask about their work.

The Inspector also handles major cases such as murder, attempt to murder and fatal accidents, with several officers assisting in investigations.

Problems identified from the Inspector interview

1. Monitoring the work of multiple officers.
2. Following up on completed and pending work.
3. Managing information related to multiple cases.
4. Handling major cases that require coordination with several officers.
5. Checking information and following up can require repeated communication.
6. Existing police software such as CCTNS has restricted access based on authorised users.

Important evidence

The Inspector described the use of a duty roster and repeatedly asking officers about their work to determine what has been completed and what remains pending.

PROBLEM IDENTIFICATION:-

After analysing the interview findings, two major problem areas were identified.

Problem Area 1 – Complaint Documentation

The Writer spends significant time manually collecting information and preparing complaint documentation. When a complainant cannot write the complaint properly, the Writer has to ask questions and prepare it manually.

Problem Area 2 – Work and Case Monitoring

The Inspector needs to monitor multiple officers, their work and pending activities. The current process involves checking duty rosters and asking officers about their work.

MOST IMPORTANT PROBLEM SELECTED:-

Selected Problem

Police station Writers spend significant time manually collecting, writing and typing complaint information, especially when complainants have difficulty writing their complaints.

This problem was selected because the Writer directly experiences the workload during complaint documentation, and the interview indicated that complaint-related writing can take about an hour.

PROPOSED SOLUTION:-

Proposed MVP Name

"AI-Assisted Police Complaint Documentation System"

The proposed system will help the Writer create a structured complaint from the complainant's spoken statement.

The important principle of the system is:

AI can extract and organize information, but it should not invent information.

The Writer will always review and approve the final complaint.

MVP FEATURES:-

Feature 1 – Complaint Registration

The Writer can enter basic complainant information such as:

- Name
- Contact number
- Date
- Time
- Location
- Complaint statement

Feature 2 – Voice-Based Complaint Input

Instead of requiring the complainant to type the complaint, the system will allow the complaint to be provided through voice.

Purpose: Reduce the difficulty faced by complainants who cannot easily write their complaints.

Feature 3 – Speech-to-Text

The spoken complaint will be converted into text.

Example:

Voice:

"My mobile phone was taken from my bag while I was travelling on the bus."

System output:

My mobile phone was taken from my bag while I was travelling on the bus.

The original transcript will be retained as the **source of truth**.

Feature 4 – Information Extraction

The system will identify information explicitly available in the complaint, such as:

- Date

- Time
- Location
- Incident
- People mentioned
- Objects mentioned
- Important keywords

The system should **not assume information that the complainant did not provide**.

Feature 5 – Complaint Draft Generation

The extracted information will be organized into a standard complaint format.

For example:

COMPLAINT DETAILS

Complainant Name:
Contact Number:
Date:
Time:
Location:

Complaint Statement:
[Original statement]

Detected Incident:
[Based only on available information]

Feature 6 – Writer Review and Editing

The Writer can review the generated draft.

The Writer can:

- Check the information
- Correct errors
- Edit the draft

- Verify it against the original statement

Feature 7 – Writer Approval

The complaint will not be considered final automatically.

The Writer must verify the information and approve the complaint.

Generated Draft

↓

Writer Verification

↓

Approve

INCLUDED IN THE MVP:-

Included

✅ Complaint registration
✅ Voice complaint input
✅ Speech-to-text
✅ Basic information extraction
✅ Structured complaint draft
✅ Writer review
✅ Writer editing
✅ Writer approval
✅ Original complaint preservation

EXPECTED BENEFIT:-

The proposed MVP aims to:

1. Reduce the Writer's manual complaint-writing workload.
2. Help complainants who have difficulty writing complaints.
3. Reduce unnecessary rewriting and typing.
4. Organize complaint information clearly.
5. Preserve the complainant's original statement.
6. Keep the Writer involved in verification and final approval.
7. Reduce the risk of unsupported information being added by the system.

FUTURE ENHANCEMENTS:-

After validating the MVP, the system can later be expanded with:

- Tamil speech-to-text
- Multilingual complaint support
- Inspector dashboard
- Officer assignment
- Pending-work tracking
- Case tracking
- Case search
- Court-date tracking
- Role-based access
- Integration with authorised police systems

These are **future features**, not part of the initial MVP.

AUDIO PROOF:-

1) Writer Interview Audio Link

https://drive.google.com/file/d/18nCzSSK6CGIq9UPXYDEw4ZaK9lBFQGDF/view?usp=drivesdk

2) Inspector Interview Audio Link

https://drive.google.com/file/d/18YkV5N7HM5d3iPP4eAFyJvNlp5wTMcJn/view?usp=drivesdk